CoreS3

SKU:K128





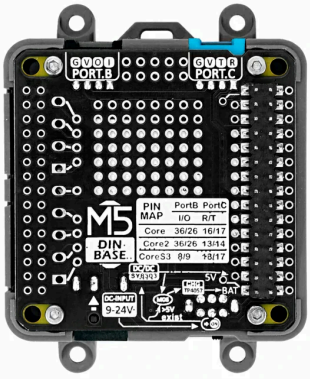


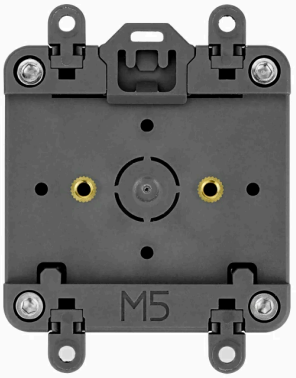














Description

CoreS3 is the third-generation main controller of the M5Stack development kit series, based on the **ESP32-S3** solution featuring a dual-core Xtensa LX7 processor running at 240 MHz with built-in Wi-Fi. It integrates 16MB Flash and 8MB PSRAM. Programs can be downloaded through the USB Type-C port, and **OTG & CDC** functions are supported for connecting USB devices and flashing firmware.

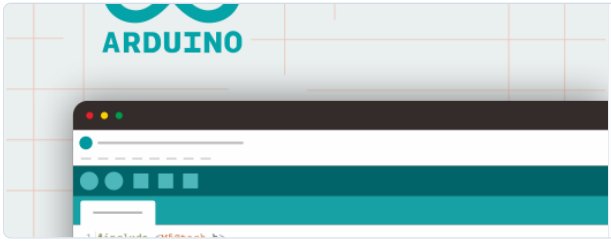
The front houses a 2.0-inch capacitive-touch IPS panel protected by high-strength glass. Below the screen is a 0.3MP GC0308 camera paired with an LTR-553ALS-WA proximity sensor. Power is managed by an AXP2101 PMU and four power-path control circuits, delivering a low-power design. Onboard sensors include a six-axis IMU BMI270 and a magnetometer BMM150. Additional peripherals comprise a microSD slot and a BM8563 RTC, enabling precise time-keeping plus sleep/timed-wake-up functions.

For audio, a 16-bit I2S amplifier AW88298 drives the built-in 1W speaker, while an ES7210 audio codec provides dual-microphone input. Independent POWER and RESET (RST) buttons are located on the side; long-pressing the RESET button enters download mode via a self-built delay circuit.

The kit ships with a DinBase for DIN-rail, wall-mount or screw mounting. It can be powered by an external DC 12V (9 ~ 24V) supply or an internal 500mAh battery. Multiple proto areas on the DinBase ease DIY expansion.

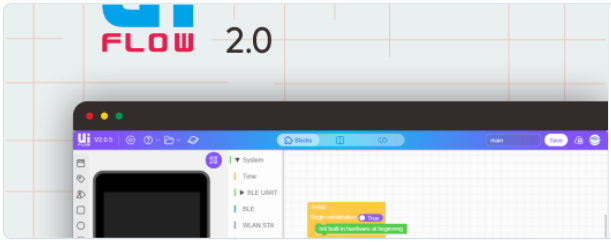
Ideal for IoT development, DIY projects, smart-home control systems and industrial automation control systems.

Tutorial



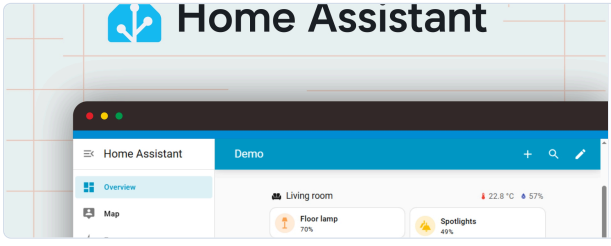
Arduino IDE

This tutorial explains how to program and control CoreS3 using Arduino IDE.



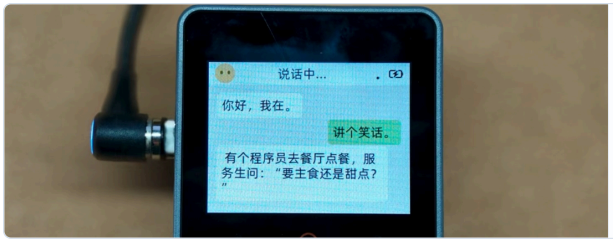
UiFlow2

This tutorial shows how to control CoreS3 via the UiFlow2 graphical programming platform.



Home Assistant

This tutorial describes how to connect CoreS3 to Home Assistant.



CoreS3 Xiaoling Voice Assistant

This tutorial will introduce how to use the CoreS3 controller to burn the Xiaoling Voice Assistant firmware via M5Burner and build a personal voice assistant application.



CoreS3 ESP-Claw Firmware Flashing

This tutorial explains how to flash the ESP-Claw firmware onto the CoreS3, enabling users to quickly configure their CoreS3 as an intelligent terminal capable of AI interaction, hardware programming, and automation control.

Features

- ESP32-based development, Wi-Fi, 16 MB Flash, 8 MB PSRAM
- Built-in camera, proximity sensor, speaker, power-indicator LED, RTC, I2S amplifier, dual microphones, capacitive touch screen, power button, reset button, gyroscope
- microSD slot
- High-strength glass panel
- Supports OTG & CDC functions
- AXP2101 power management, low-power design
- Development Platform
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 × CoreS3
- 1 × DinBase
- 1 × Hex Key L-Shape 2.5 mm (For M3 Screw)
- 4 × Screw Buckle
- 1 × Rail-mount Base Buckle

Applications

- IoT development
- DIY project development
- Smart-home control systems
- Industrial automation control systems

Specifications

Specification	Parameter
SoC	ESP32-S3 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz
Flash	16MB
PSRAM	8MB Quad
Wi-Fi	2.4 GHz Wi-Fi
Touch IPS LCD	2.0" @320 × 240 ILI9342C / ILI9342E
Camera	GC0308 0.3MP
Proximity Sensor	LTR-553ALS-WA
Power Management	AXP2101
6-axis IMU	BMI270
3-axis Magnetometer	BMM150
RTC	BM8563
Speaker	16-bit I2S amplifier AW88298 @1W
Audio Codec	ES7210, dual-microphone input
Product Size	Whole set (CoreS3+DinBase): 69.0 × 54.0 × 31.5mm Main unit (CoreS3): 54.0 × 54.0 × 15.5mm
Product Weight	72.7g
Package Size	105.6 × 66.0 × 35.3mm
Gross Weight	101.8g

Learn

BMM150 Magnetic Interference

Products containing magnets may interfere with the BMM150 magnetic sensor, resulting in abnormal readings. When used with M5 controllers that have magnets, remove the magnets and avoid placing the BMM150 near strong magnetic fields.

Download Mode

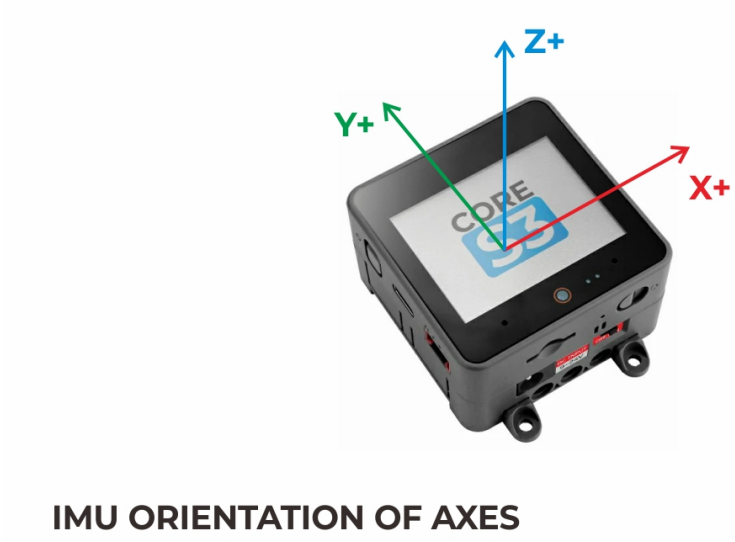
Press and hold the RESET button for 3 seconds. When the green indicator LED turns on, release the button. After release, the green indicator LED turns off, indicating that the device has entered download mode.



Power On/Off

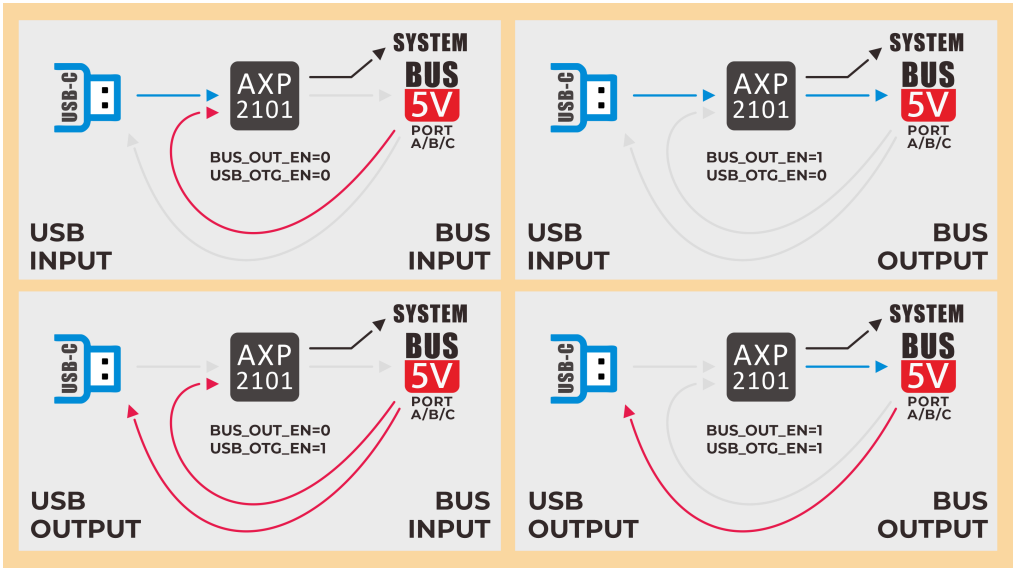
- Power-on: single-click the left POWER button
- Power-off: long-press the left POWER button for 6 s
- Reset: single-click the bottom RST button

IMU Triaxial Direction Schematic Diagram



Power Management

CoreS3 uses the AXP2101 PMU together with the AW9523B IO expander to control power input/output directions. Refer to the pin states of `BUS_OUT_EN` and `USB_OTG_EN` in the figure below and check the example code in CoreS3 Power Manager Example。

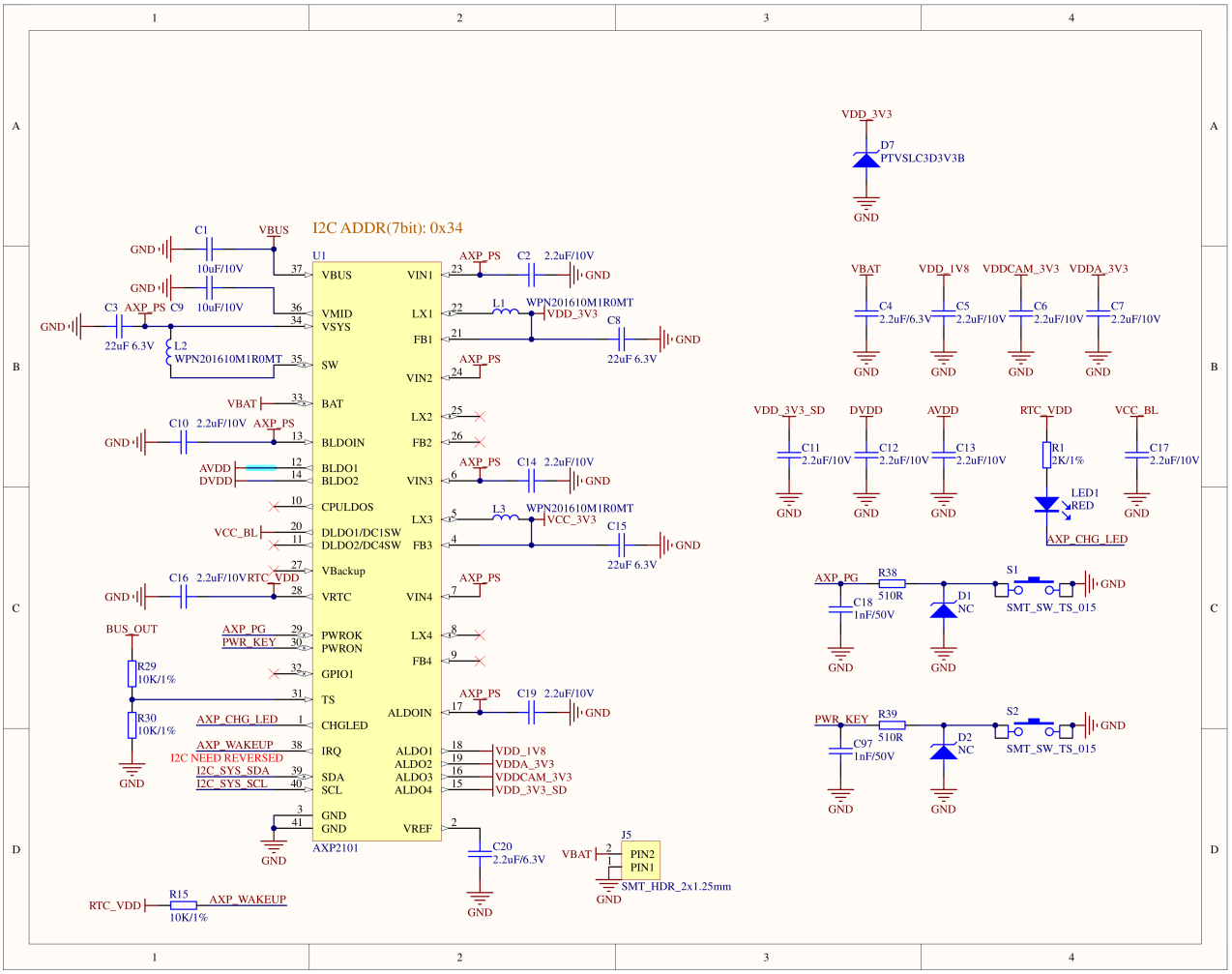


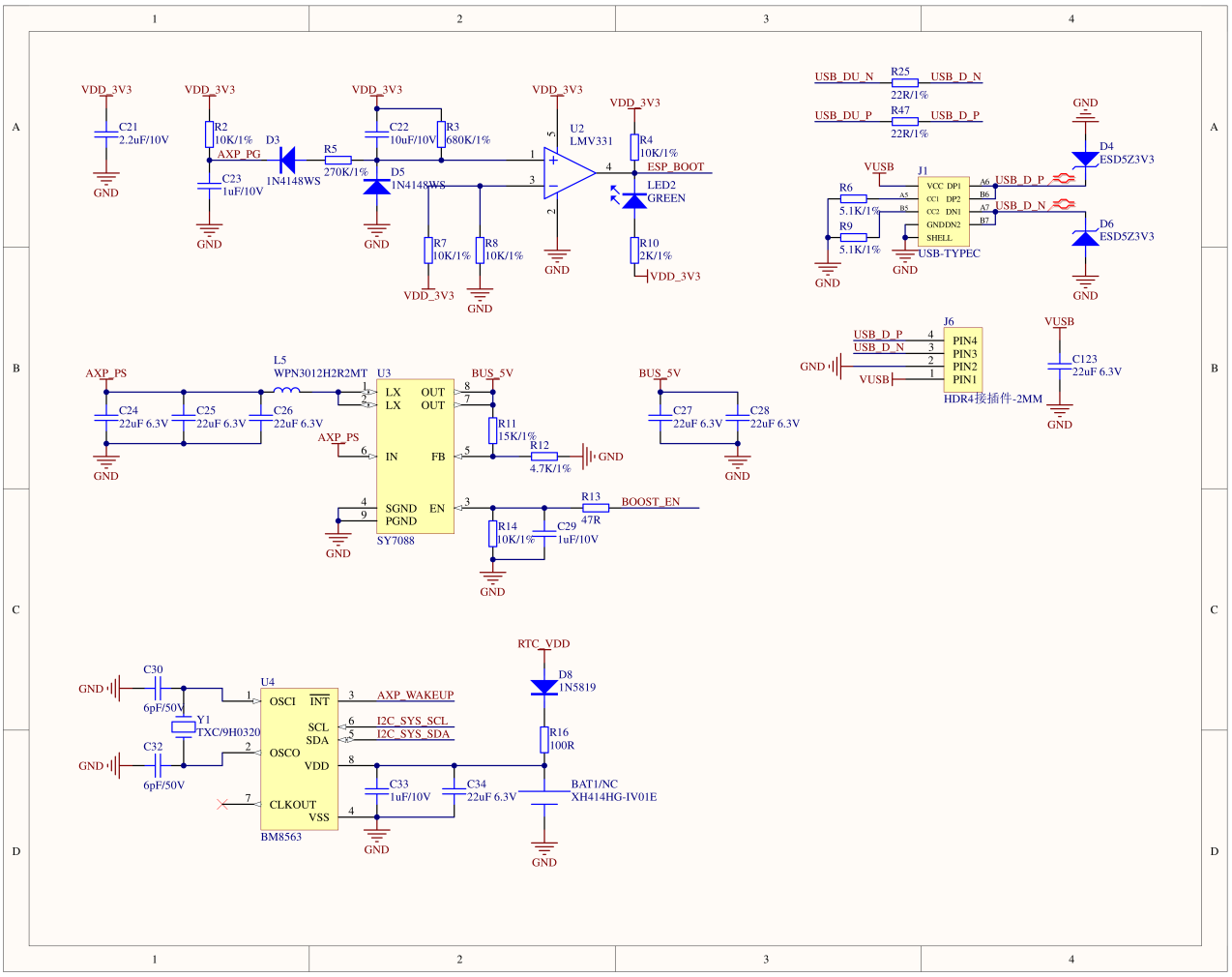
Certifications

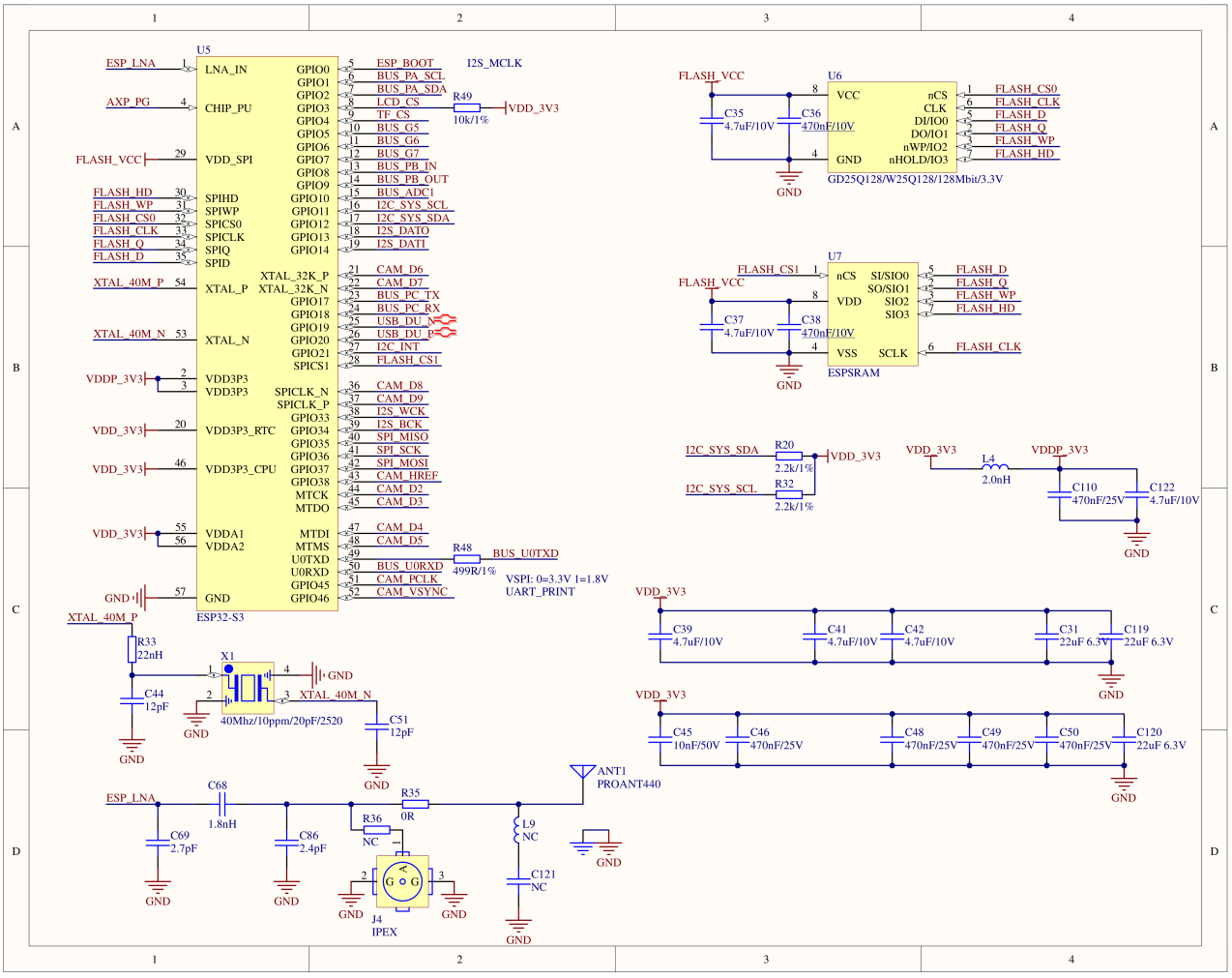
- CE/MIC/FCC/SAR

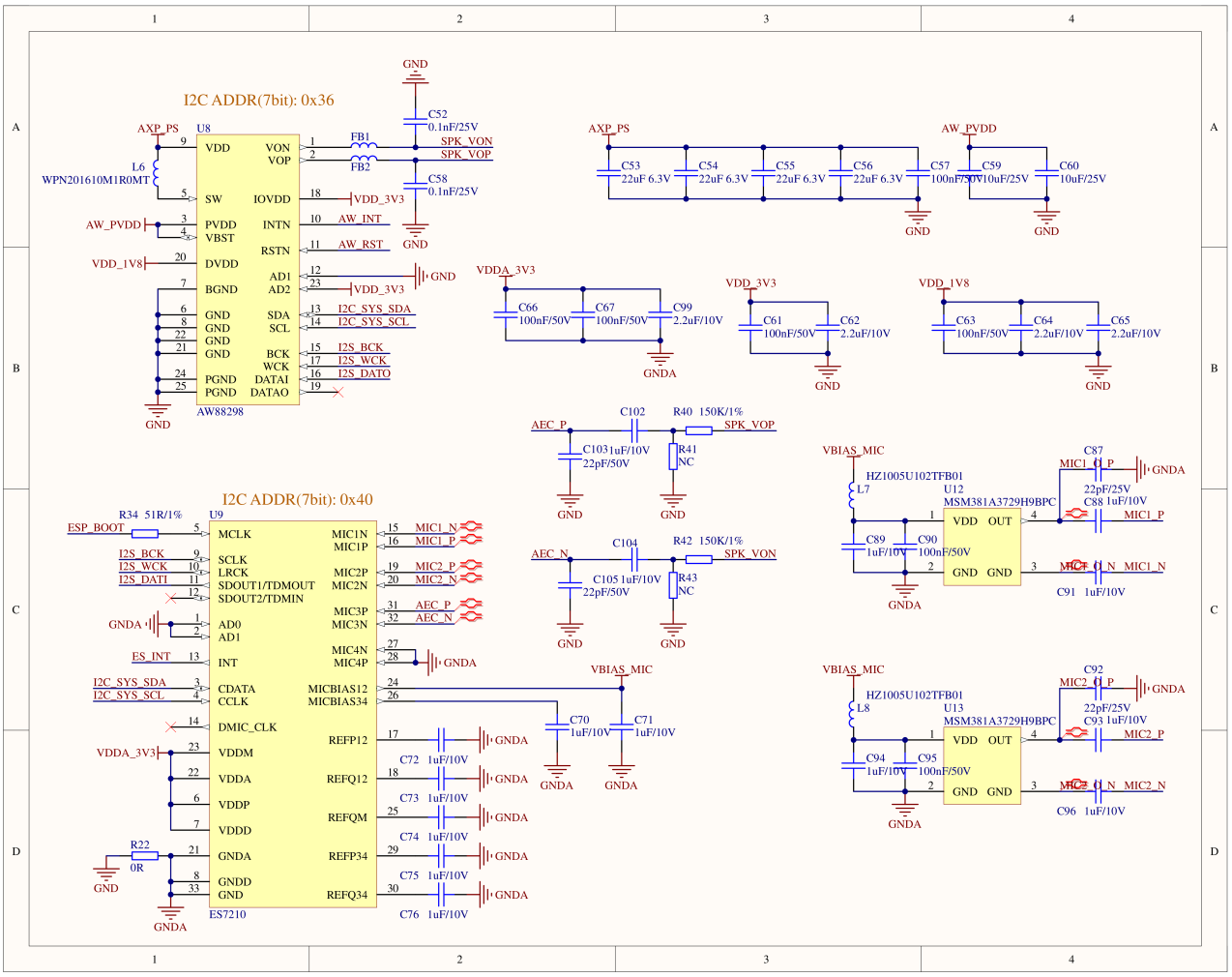
Schematics

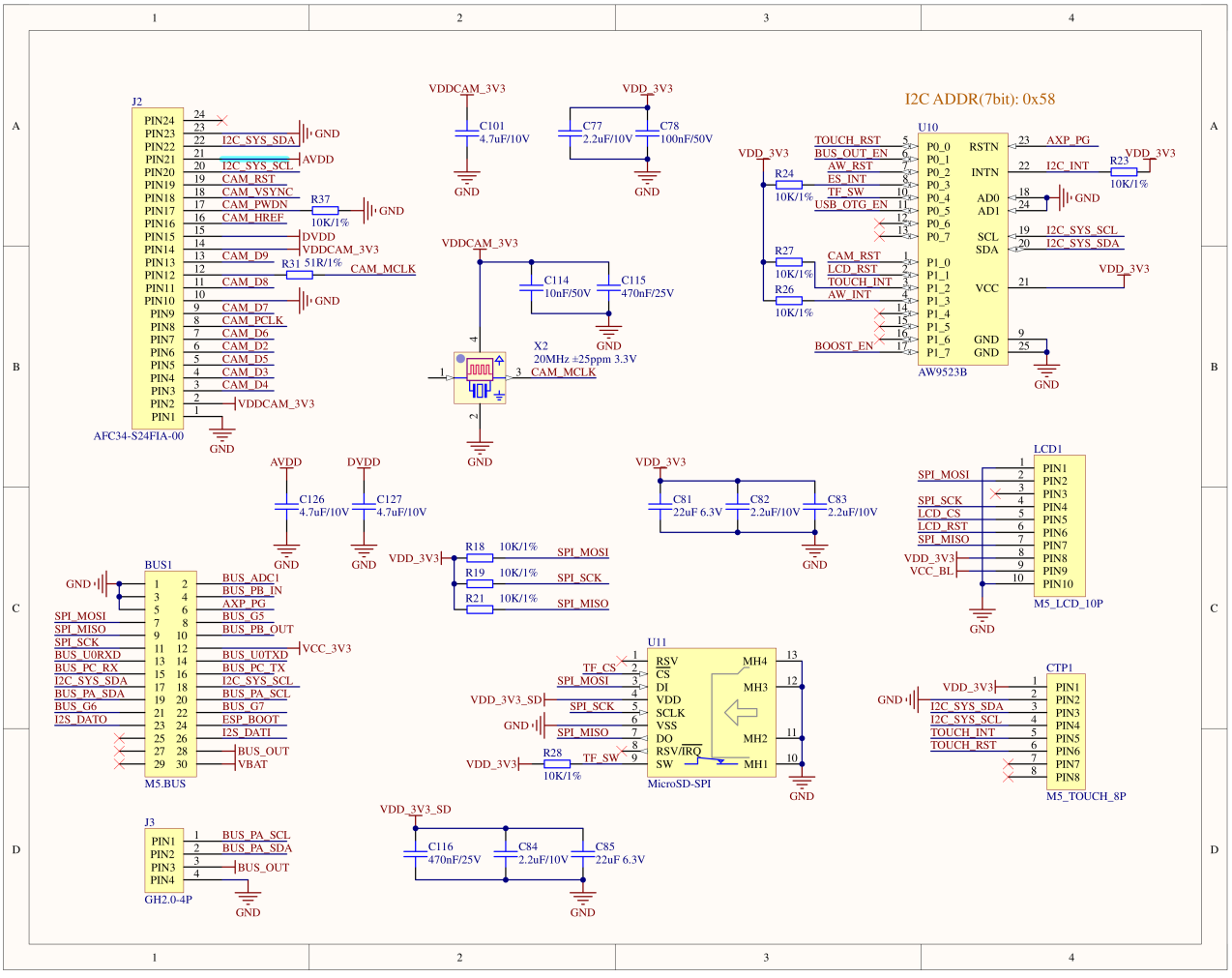
- CoreS3 Schematics PDF
- Base DIN Schematics PDF

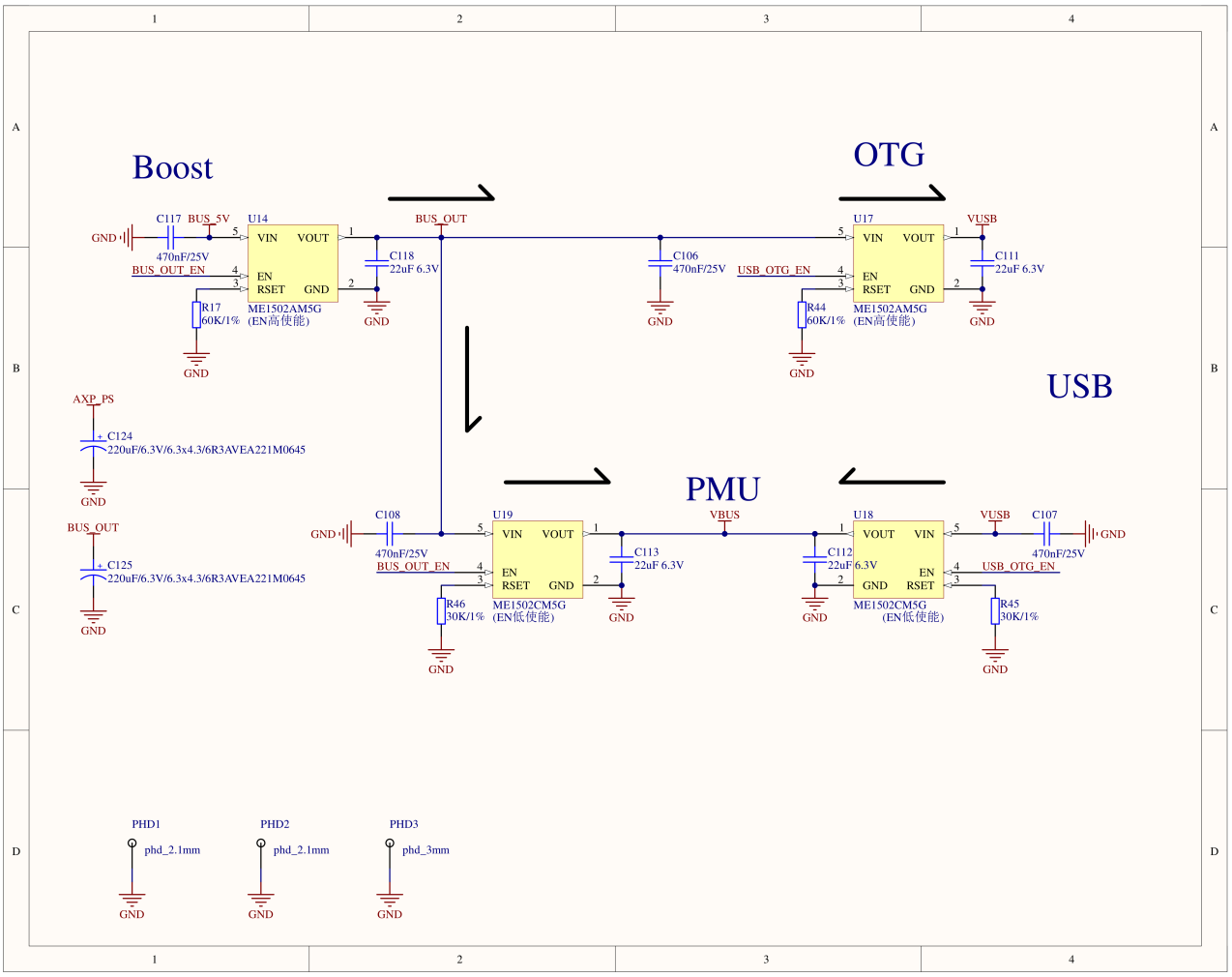


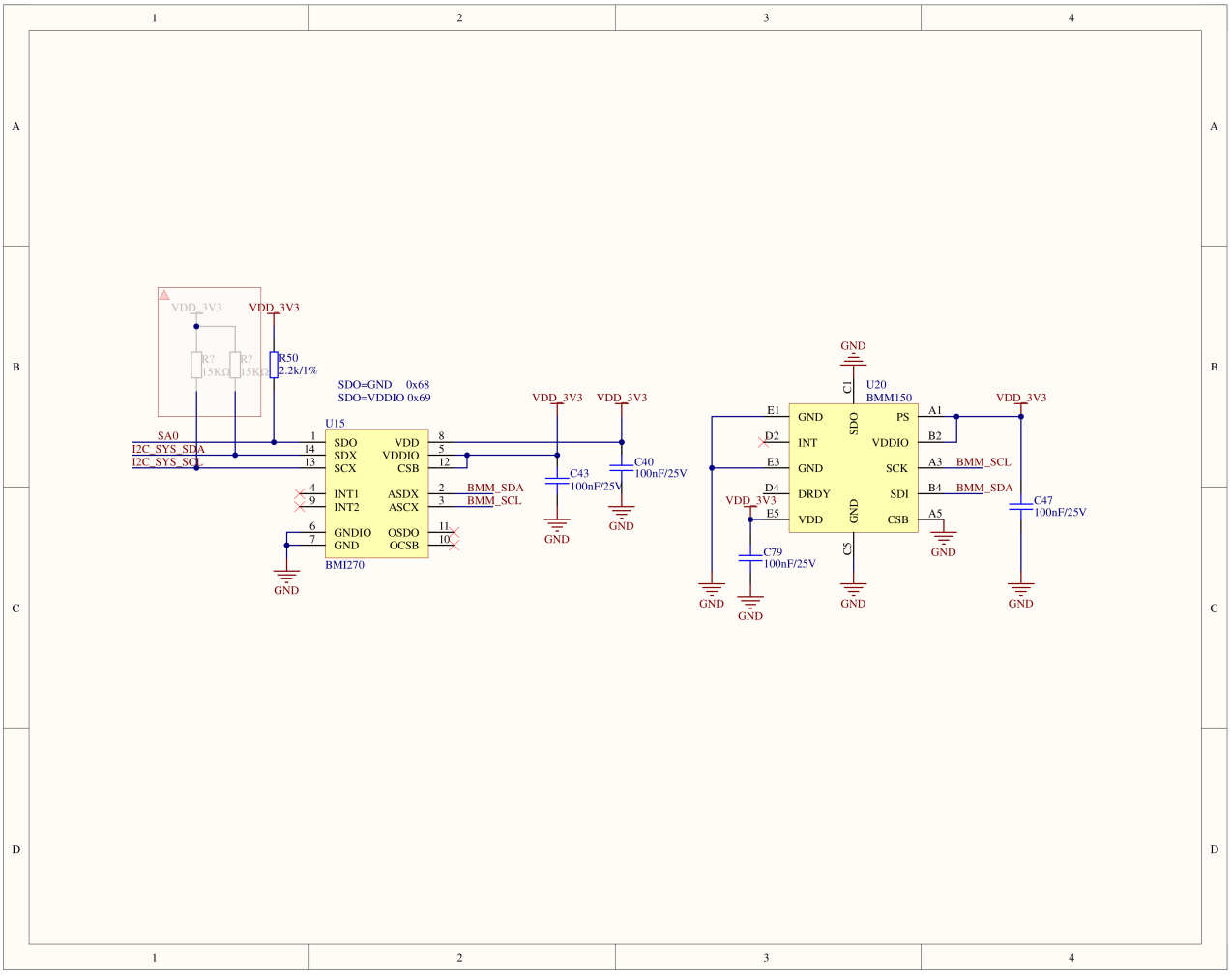

VDD_3V3
C21
2.2uF/10V
GND
R2
10K/1%
AXP_PG
C23
1uF/10V
D3
1N4148WS
R5
270K/1%
C22
10uF/10V
R3
680K/1%
D5
1N4148WS
U2
LMV331
R4
10K/1%
ESP_BOOT
LED2
GREEN
R10
2K/1%
R7
10K/1%
R8
10K/1%
USB_DU_N
R25
22R/1%
USB_D_N
USB_DU_P
R47
22R/1%
USB_D_P
VUSB
J1
R6
5.1K/1%
R9
5.1K/1%
USB-TYPEC
D4
ESD5Z3V3
D6
ESD5Z3V3
J6
HDR4接插件-2MM
C123
22uF 6.3V
AXP_PS
L5
WPN3012H2R2MT
U3
SY7088
BUS_5V
C24
22uF 6.3V
C25
22uF 6.3V
C26
22uF 6.3V
R11
15K/1%
R12
4.7K/1%
C27
22uF 6.3V
C28
22uF 6.3V
R13
47R
BOOST_EN
R14
10K/1%
C29
1uF/10V
C30
6pF/50V
Y1
TXC/9H0320
C32
6pF/50V
U4
BM8563
AXP_WAKEUP
I2C_SYS_SCL
I2C_SYS_SDA
RTC_VDD
D8
1N5819
R16
100R
C33
1uF/10V
C34
22uF 6.3V
BAT1/NC
XH414HG-IV01E

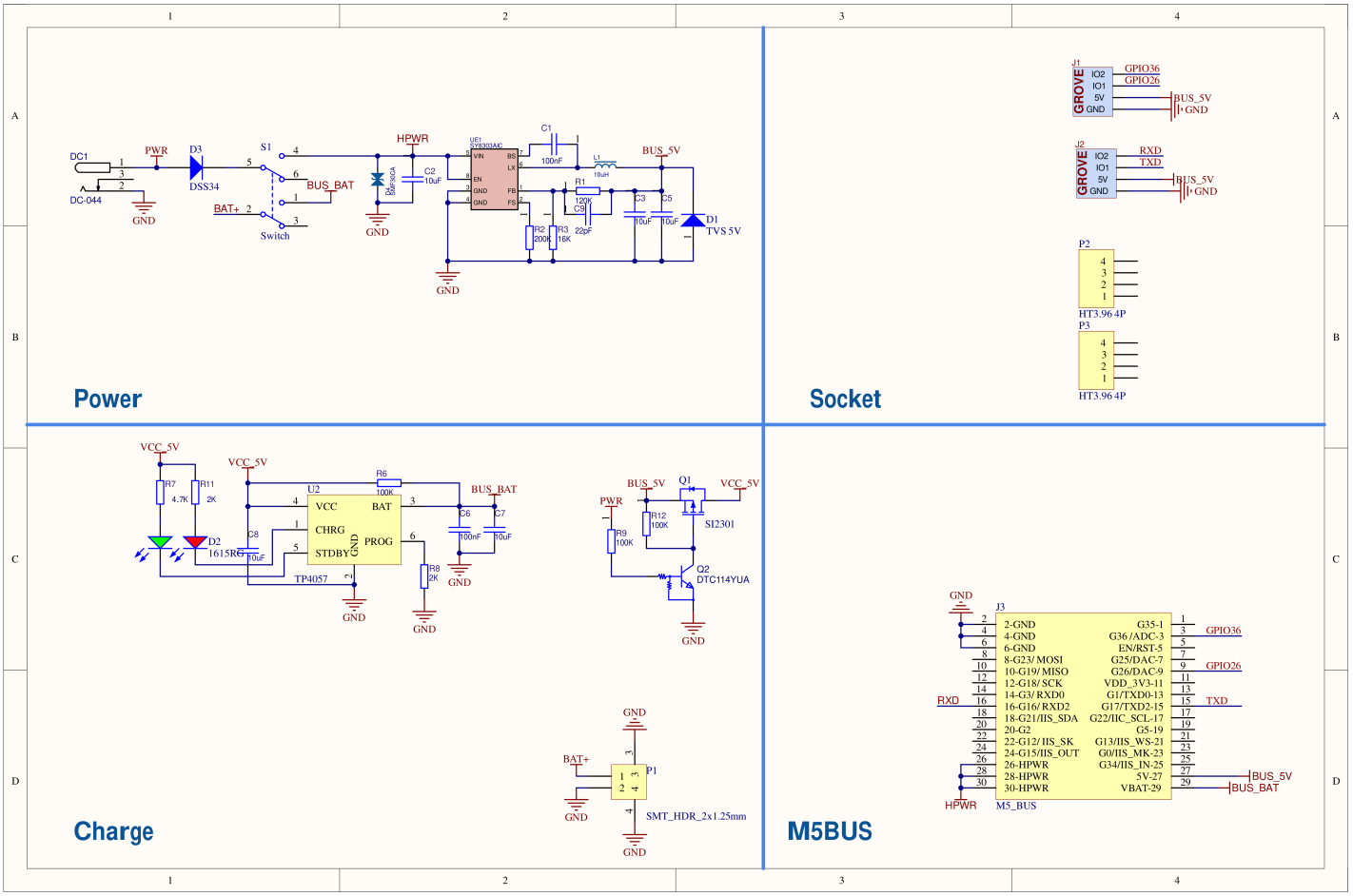

U5
ESP32-S3
ESP_LNA
AXP_PG
FLASH_VCC
VDDP_3V3
VDD_3V3
XTAL_40M_P
XTAL_40M_N
GND
I2S_MCLK
R49
10k/1%
R48
499R/1%
VSPI: 0=3.3V 1=1.8V
UART_PRINT
U6
GD25Q128/W25Q128/128Mbit/3.3V
U7
ESPSRAM
FLASH_CS0
FLASH_CLK
FLASH_D
FLASH_Q
FLASH_WP
FLASH_HD
FLASH_CS1
C35
4.7uF/10V
C36
470nF/10V
C37
4.7uF/10V
C38
470nF/10V
I2C_SYS_SDA
I2C_SYS_SCL
R20
2.2k/1%
R32
2.2k/1%
L4
2.0nH
C110
470nF/25V
C122
4.7uF/10V
C39
4.7uF/10V
C41
4.7uF/10V
C42
4.7uF/10V
C31
22uF 6.3V
C119
22uF 6.3V
C45
10nF/50V
C46
470nF/25V
C48
470nF/25V
C49
470nF/25V
C50
470nF/25V
C120
22uF 6.3V
R33
22nH
X1
40Mhz/10ppm/20pF/2520
C44
12pF
C51
12pF
C68
1.8nH
C69
2.7pF
C86
2.4pF
R35
0R
R36
NC
L9
NC
C121
NC
ANT1
PROANT440
J4
IPEX


I2C ADDR(7bit): 0x36
AW88298
I2C ADDR(7bit): 0x40
ES7210
AXP_PS
AW_PVDD
VDDA_3V3
VDD_3V3
VDD_1V8
GND
GNDA
SPK_VON
SPK_VOP
AEC_P
AEC_N
VBIAS_MIC
HZ1005U102TFB01
MSM381A3729H9BPC
MIC1_P
MIC1_N
MIC2_P
MIC2_N


I2C ADDR(7bit): 0x58
AW9523B
AFC34-S24FIA-00
M5_LCD_10P
M5_TOUCH_8P
M5.BUS
MicroSD-SPI
VDDCAM_3V3
VDD_3V3
VDD_3V3_SD
VCC_3V3
VBAT
CAM_MCLK
I2C_SYS_SDA
I2C_SYS_SCL
SPI_MOSI
SPI_SCK
SPI_MISO


Boost
OTG
USB
PMU
C117 BUS_5V
470nF/25V
BUS_OUT_EN
U14
VIN VOUT
EN
RSET GND
ME1502AM5G
(EN高使能)
R17
60K/1%
BUS_OUT
C118
22uF 6.3V
C106
470nF/25V
U17
USB_OTG_EN
R44
60K/1%
VUSB
C111
22uF 6.3V
AXP_PS
C124
220uF/6.3V/6.3x4.3/6R3AVEA221M0645
C125
220uF/6.3V/6.3x4.3/6R3AVEA221M0645
C108
470nF/25V
U19
ME1502CM5G
(EN低使能)
R46
30K/1%
C113
22uF 6.3V
VBUS
U18
C112
22uF 6.3V
C107
470nF/25V
R45
30K/1%
PHD1
phd_2.1mm
PHD2
phd_2.1mm
PHD3
phd_3mm
GND


VDD_3V3
R50
2.2k/1%
SDO=GND 0x68
SDO=VDDIO 0x69
U15
SA0
I2C_SYS_SDA
I2C_SYS_SCL
SDO VDD
SDX VDDIO
SCX CSB
INT1 ASDX
INT2 ASCX
GNDIO OSDO
GND OCSB
BMI270
BMM_SDA
BMM_SCL
C43
100nF/25V
C40
100nF/25V
U20
BMM150
GND
INT
DRDY
VDD
SDO
VDDIO
SCK
SDI
CSB
PS
C79
100nF/25V
C47
100nF/25V
C1
C5



PinMap

LCD Screen

LCD Pixels: 320x240

ESP32-S3	G37	G36	G3	G35
ILI9342C / ILI9342E	MOSI	SCK	CS	DC

AW9523B (0x58)	P1_1
ILI9342C / ILI9342E	RST

AXP2101 (0x34)	DLDO1	LX1
ILI9342C / ILI9342E	BL	PWR

microSD

TF Card Maximum Supported Capacity: 16GB

ESP32-S3	G35	G37	G36	G4
TF Card	MISO	MOSI	SCK	CS

Camera & Proximity Sensor LTR-553ALS-WA

ESP32-S3	G12	G11	G45	G46	G38
GC0308 (0x21)	I2C_SYS_SDA	I2C_SYS_SCL	CAM_PCLK	CAM_VSYNC	CAM_HREF
LTR-553ALS-WA (0x23)	I2C_SYS_SDA	I2C_SYS_SCL			

AW9523B	P1_0
GC0308	CAM_RST

> The LTR-553ALS-WA proximity sensor and the camera are integrated on one ribbon cable, and use I2C for communication. Please refer to the table above for detailed communication addresses.

GC0308

Interface	Camera Pin	ESP32-S3
SCCB Clock	SIOC	G11
SCCB Data	SIOD	G12
System Clock	XCLK	-1
Vertical Sync	VSYNC	G46
Horizontal Reference	HREF	G38
Pixel Clock	PCLK	G45
Pixel Data Bit 0	D0	G39
Pixel Data Bit 1	D1	G40
Pixel Data Bit 2	D2	G41
Pixel Data Bit 3	D3	G42
Pixel Data Bit 4	D4	G15
Pixel Data Bit 5	D5	G16
Pixel Data Bit 6	D6	G48
Pixel Data Bit 7	D7	G47
Camera Reset	RESET	-1
Camera Power Down	PWDN	-1

CAP.TOUCH

ESP32-S3	G12	G11
FT6336U (0x38)	I2C_SYS_SDA	I2C_SYS_SCL

AW9523B	P0_0	P1_2
FT6336U	TOUCH_RST	TOUCH_INT

Microphone & Amplifier

ESP32-S3	G12	G11	G34	G33	G13	G14	G0
ES7210 (0x40)	I2C_SYS_SDA	I2C_SYS_SCL	I2S_BCK	I2S_WCK		I2S_DATI	I2S_MCLK
AW88298 (0x36)	I2C_SYS_SDA	I2C_SYS_SCL	I2S_BCK	I2S_WCK	I2S_DATO		

AW9523B	P0_2	P1_3
AW88298	AW_RST	AW_INT

AXP Power Indicator Light

AXP2101	AXP_CHG_LED
Red LED	RTC_VDD

RTC

ESP32-S3	G12	G11
BM8563 (0x51)	I2C_SYS_SDA	I2C_SYS_SCL

AXP2101	IRQ
BM8563	AXP_WAKEUP

IMU (3-axis Gyroscope + 3-axis Accelerometer + 3-axis Magnetometer)

ESP32-S3	G12	G11
BMI270 (0x69)	I2C_SYS_SDA	I2C_SYS_SCL

Internal I2C Connection

ESP32-S3	G12	G11
BMI270	I2C_SYS_SDA	I2C_SYS_SCL
AXP2101	I2C_SYS_SDA	I2C_SYS_SCL
BM8563	I2C_SYS_SDA	I2C_SYS_SCL
ES7210	I2C_SYS_SDA	I2C_SYS_SCL
AW88298	I2C_SYS_SDA	I2C_SYS_SCL

BMM150

BMI270	BMI270_ASDx	BMI270_ASCx
BMM150 (0x10)	BMM_SDA	BMM_SCL

BMM150 Mounted on BMI270

Connected through BMI270's Sensor Hub auxiliary I2C interface to access BMM150 and achieve unified 9-axis sensor data acquisition.

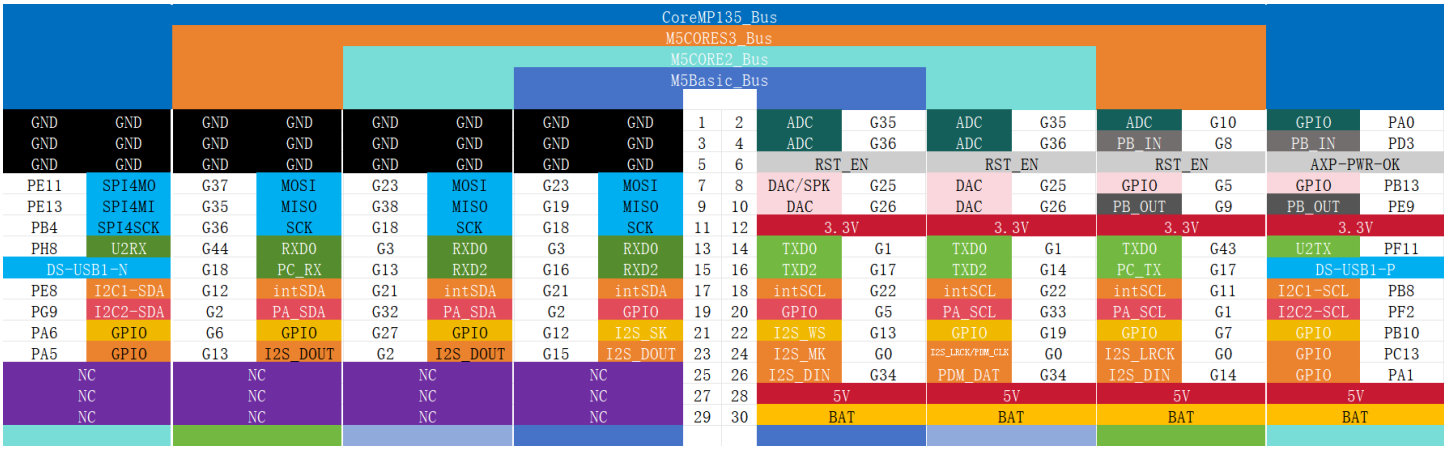
HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	G2	G1
PORT.B	GND	5V	G9	G8
PORT.C	GND	5V	G17	G18

CoreS3 M5-Bus Diagram

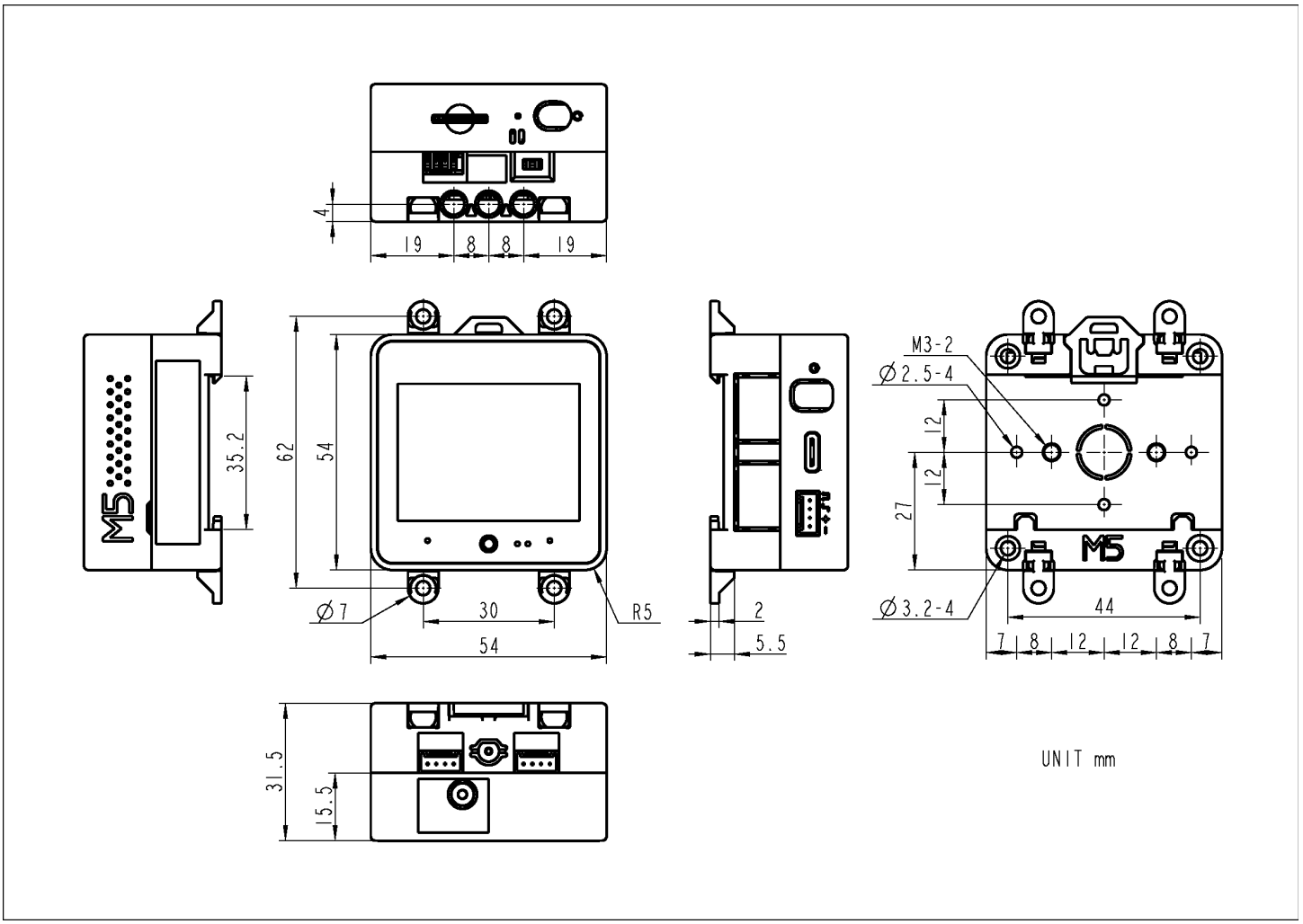
FUNC	PIN	LEFT	RIGHT	PIN	FUNC
	GND	1	2	G10	ADC
	GND	3	4	G8	PB_IN
	GND	5	6	RST	EN
MOSI	G37	7	8	G5	GPIO
MISO	G35	9	10	G9	PB_OUT
SCK	G36	11	12	3V3	
RXD0	G44	13	14	G43	TXD0
PC_RX	G18	15	16	G17	PC_TX
Int SDA	G12	17	18	G11	Int SCL
PORT.A SDA	G2	19	20	G1	PORT.A SCL
GPIO	G6	21	22	G7	GPIO
I2S_DOUT	G13	23	24	G0	I2S_LRCK
	HVIN(Base DIN)	25	26	G14	I2S_DIN
	HVIN(Base DIN)	27	28	5V	
	HVIN(Base DIN)	29	30	BAT	

Core Series PinMap Comparison



Model Size

- Cores3 Model Size DXF



PCB

- Cores3 PcbDoc

Description

This file can be used for DIY adaptation of CoreS3 Module Type A expansion modules.

Structure

- CoreS3 Structure Files

Datasheets

- ESP32-S3
- LTR-553ALS-WA
- GC0308
- ES7210
- BMM150
- BMI270
- BM8563
- AXP2101
- AW88298
- AW9523B

- ILI9342E

Softwares

Quick Start

- CoreS3 OpenAI Voice Assistant
- CoreS3 XiaoZhi Voice Assistant
- CoreS3 HA Voice Assistant
- CoreS3 ESP-Claw Firmware Flashing

Arduino

- CoreS3 Arduino Quick Start
- CoreS3 Arduino Library
- CoreS3 Arduino M5Unified Library
- CoreS3 Arduino M5GFX Library

UiFlow2

- CoreS3 UiFlow2 Quick Start
- CoreS3 UiFlow2 Book

PlatformIO

- CoreS3 Factory Firmware

```
[env:m5stack-cores3]
platform = espressif32@6.7.0
board = esp32-s3-devkitc-1
framework = arduino
upload_speed = 1500000
monitor_speed = 115200
build_flags =
    -DESP32S3
    -DBOARD_HAS_PSRAM
    -mfix-esp32-psram-cache-issue
    -DCORE_DEBUG_LEVEL=5
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DARDUINO_USB_MODE=1
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

ESP-IDF

- Espressif's Board Support Packages – CoreS3
- CoreS3 ESP-IDF BSP Guide

Easyloader

Easyloader	Download	Note
CoreS3 Factory Firmware Easyloader	download	/

Flash CoreS3 factory firmware with Web Tools

Other

- CoreS3 Restore Factory Firmware Guide
- I2C Address Table

Chip	Address
AW88298	0x36
AW9523	0x58
AXP2101	0x34
BM8563	0x51
BMI270	0x69
BMM150	0x10
ES7210	0x40
FT6336	0x38
GC0308	0x21
LTR553	0x23

Video

Product Comparison

Hardware Peripheral			
	CoreS3-Lite	CoreS3	CoreS3-SE
Camera (GC0308)	✅	✅	❌
Proximity Sensor (LTR-553ALS-WA)	✅	✅	❌
IMU (BMI270)	✅	✅	❌
Compass (BMM150)	✅	✅	❌
RTC	✅	✅	✅
Microphone	✅	✅	✅
Speaker	✅	✅	✅
PMIC (AXP2101)	✅	✅	✅
16 MB Flash & 8 MB PSRAM	✅	✅	✅
Touch	✅	✅	✅
Back Cover	Cover For CoreS3	Base DIN	❌
Battery Capacity	200 mAh	500 mAh	❌

To compare information on the controller series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.

Version Change

Release Date	Product Change	Note
2026.8.7	LCD driver changed from ILI9342C to ILI9342E	M5GFX library version >= 0.2.27 is required